

Economy

The Institute Employment Report: March 2026

08 April 2026

Key takeaways

- An estimate of payrolls growth based on Bank of America customer deposit account data continues to suggest a reacceleration in payrolls growth, with year-over-year (YoY) gains rising to 1.4% in March - back in line with early-2025 momentum.
- But while overall jobs growth has improved, after-tax wage growth is increasingly "K-shaped," with higher-income households seeing wage growth of 5.6% YoY in March, versus just 1.0% and 2.0% for lower- and middle-income groups, respectively, according to Bank of America customer account data. This is the widest gap in growth rates since 2015.
- In our view, a surge in bonuses among higher-income households is likely amplifying wage dispersion early in 2026, with solid YoY growth in bonuses for higher-income households while YoY growth in bonuses for middle- and lower-income cohorts remain negative. But even if this effect fades over this year, underlying wage dynamics continue to favor the top end.

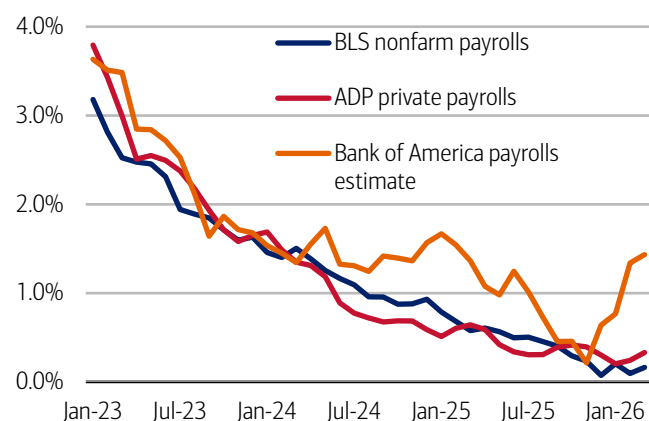
Payrolls growth showed continued momentum in March...

Bank of America deposit account data suggests the significant recovery in payrolls growth seen in February continued into March, while unemployment payments growth has eased. However, the "K" shape in wage growth grew wider.

We use Bank of America consumer deposit data to estimate a payrolls series by looking at how the number of customer accounts receiving a paycheck is changing (see Methodology). This data can be fairly noisy, partly due to seasonal variation. However, looking at the three-month moving average, Exhibit 1 suggests that the year-over-year (YoY) growth in our measure rose to 1.4% in March 2026, up from 1.3% YoY in February.

Exhibit 1: Bank of America account data indicates payrolls growth has rebounded to rates comparable to those observed in early 2025

Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



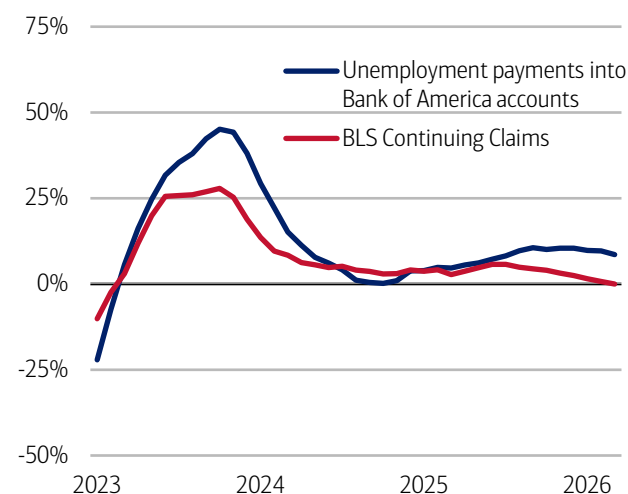
Source: Bank of America internal data, Haver Analytics

Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

BANK OF AMERICA INSTITUTE

Exhibit 2: Unemployment payments growth into Bank of America customer accounts has slowed

Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

BANK OF AMERICA INSTITUTE

The Bureau of Labor Statistics (BLS) payrolls growth figure shows considerably weaker YoY jobs growth than the estimate based on Bank of America data. In February, BLS payrolls dropped by 133K month-over-month (MoM), though the number of jobs was still up by 149K YoY. In the latest March data, the BLS estimates payrolls rose 178K MoM and are up 260K YoY, which appears directionally consistent with the stronger picture seen in Bank of America data.

Additionally, Bank of America data on unemployment payments into customer accounts shows growth has eased. In March, unemployment payments growth dropped below 9% YoY (Exhibit 2). This also appears directionally consistent with BLS Continuing Claims data, which shows YoY growth around zero. The overall household survey unemployment rate was 4.3% in March, down from 4.4% in February.

... but there is a very large gap in after-tax wage growth between higher-income households and other cohorts

While it appears jobs growth momentum has improved, there is a very large gap between after-tax wage and salary growth across income cohorts. In fact, Bank of America deposit data shows higher-income households' after-tax wage and salary growth rose to 5.6% YoY in March, up from 4.2% YoY in February. This is the highest after-tax wage growth for this cohort since August 2021 (Exhibit 3).

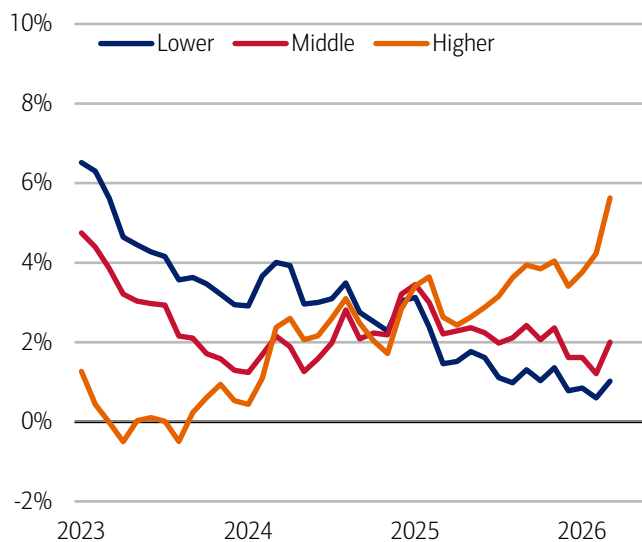
There was also a rebound in middle- and lower-income households' after-tax wage growth in March, to 2.0% and 1.0% YoY, respectively. But the gap between these cohorts' and higher-income households' wage growth still widened and is now the largest gap we've seen since the data series began in 2015.

In our view, one driver of the current widening in after-tax wage growth between higher-income households and the other cohorts is likely bonus growth. An estimate of bonus growth (see Methodology) based on Bank of America deposit data showed higher-income households saw solid positive growth over the first three months of 2026, while for middle- and lower-income households, bonus growth was negative (Exhibit 4).

This divergence in bonus growth may fade as we move through the year (many higher-income households receive their bonuses in the early part of the year). But, even so, the persistence in the "K" shape between higher-income wage growth and other cohorts suggests underlying wage dynamics may continue to favor the higher-income group.

Exhibit 3: In March, higher-income households' after-tax wage growth rose to 5.6% YoY, while lower-income households' wage growth was 1.0% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, SA)

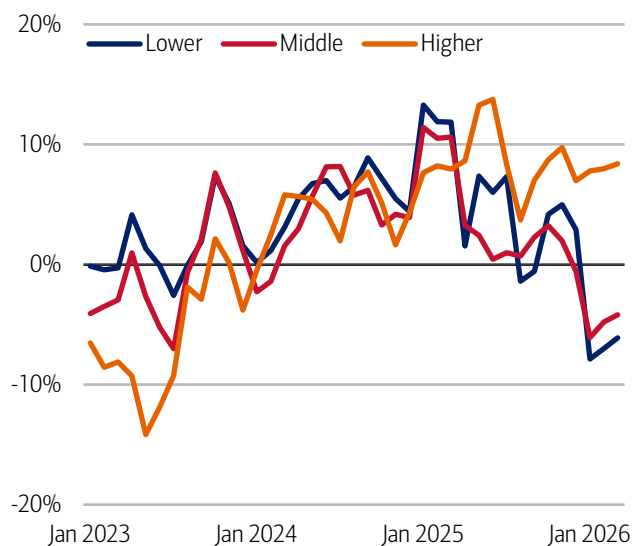


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 4: It appears there has been a large divergence in bonus growth in 2026

Estimate of average bonus payments by household income terciles (3-month moving average, YoY)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under \$5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation,

changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Estimate of payrolls growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth.

An estimate of bonus growth from Bank of America deposit data is calculated by looking at customers who have received an inbound ACH payroll transaction in the last two years. From this sample an estimate of bonuses is derived by looking for payroll transactions which are over 50% higher than the median regular payroll payments received by the customer. Of these payments only those that were received around the same time in each of the last two years are selected.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

David Michael Tinsley

Senior Economist, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Yan Peng

Vice President, Global Risk Analytics

Akshita Jain

Assistant Vice President, Global Risk Analytics

Jasmine Elder

Vice President, Global Risk Analytics

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice. Copyright 2026 Bank of America Corporation. All rights reserved.

Economy

Golf spending is slightly above par

02 April 2026

Key takeaways

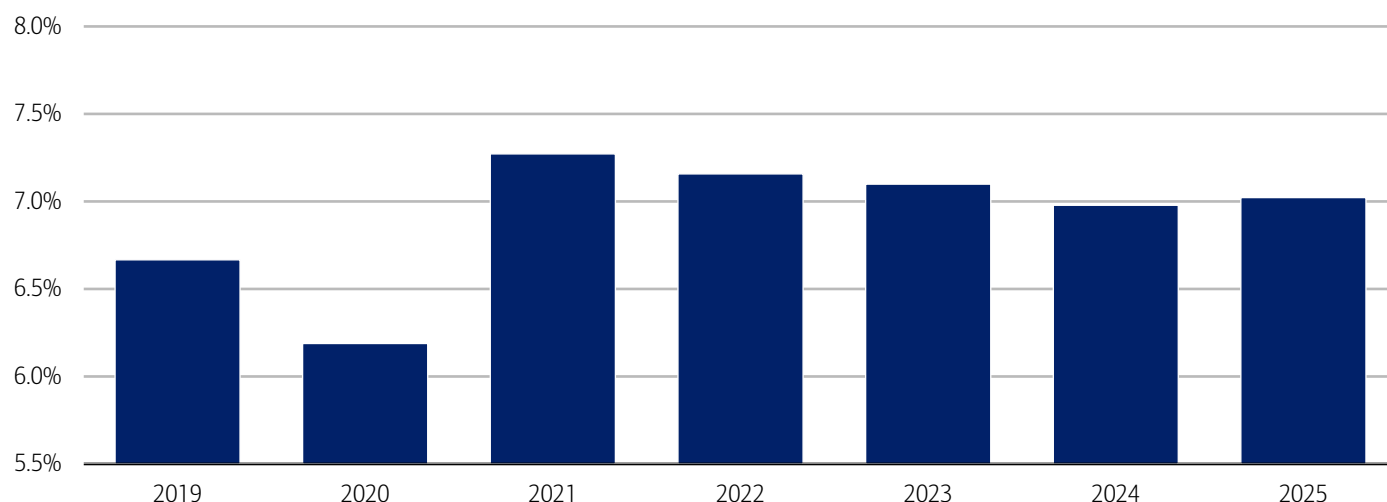
- After three years of declining golf participation, from the 2021 pandemic peak, Bank of America payments data shows a modest rebound in 2025. At the same time, average annual spend per golfer has increased for four consecutive years, suggesting a smaller but more enthusiastic group of participants.
- Gen Z has continued to gain share in golf participation, likely driven by lower-cost, more accessible formats like driving ranges and simulators. At the same time, participation among younger Millennials has declined, potentially reflecting competing life priorities and tighter budgets - though those who remain active appear to be spending more.
- Regionally, the West has recorded the strongest golf spending growth by region in 2025, supported by domestic migration, destination-style "stay-to-play" courses, and expanded offerings. Other regions are seeing new players enter the sport, but typically through lower-cost golfing options.

Golf teed up a comeback in 2025

Golf began climbing out of the rough in 2025, reversing a three-year decline in the proportion of households paying for golf activities (e.g., golf courses, driving ranges, mini golf and simulators but not country club dues), according to Bank of America aggregated credit and debit card data (Exhibit 1). And our data suggests that, despite the pullback from the pandemic peak, participation remains above 2019 levels.

Exhibit 1: The share of households with golf spending peaked in 2021 and declined until 2024 before stabilizing in 2025

Share of US households with golf spending (yearly, %)



Source: Bank of America internal data. Note: Golf spending includes spending at golf courses, mini golf, driving ranges, and simulators but does not include country club dues or golf retailers.

BANK OF AMERICA INSTITUTE

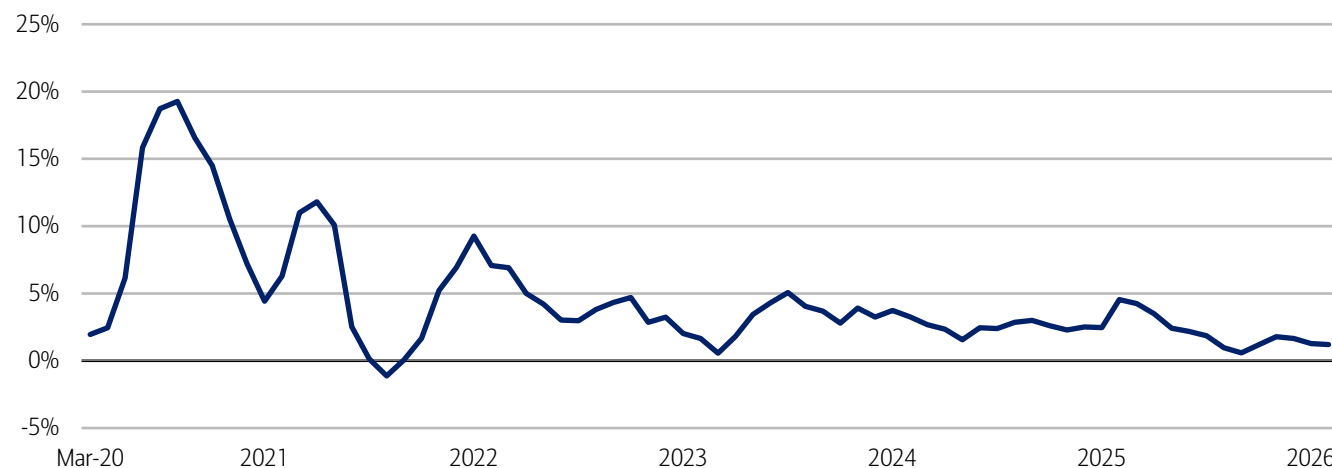
Furthermore, most people who play golf seem to be deepening their relationship with the sport. Bank of America internal data found that per household spending continued to grow over the past four years, although it slowed to around 1% year-over-year (YoY) in February 2026 (Exhibit 2).

According to the National Golf Foundation, the number of rounds played has increased for the past three years, despite a decrease in the actual number of golf courses nationwide. Golf has also potentially received a boost from the ease and

accessibility of driving ranges and indoor simulators. So overall, while the share of US households that play golf has held fairly steady over the past two years, the households that do play seem to be playing a bit more often.

Exhibit 2: Of the households that spend on golf, outlays have increased through February 2026, although at a slightly slower rate

Golf spending per household (3-month moving average, YoY%)



Source: Bank of America internal data

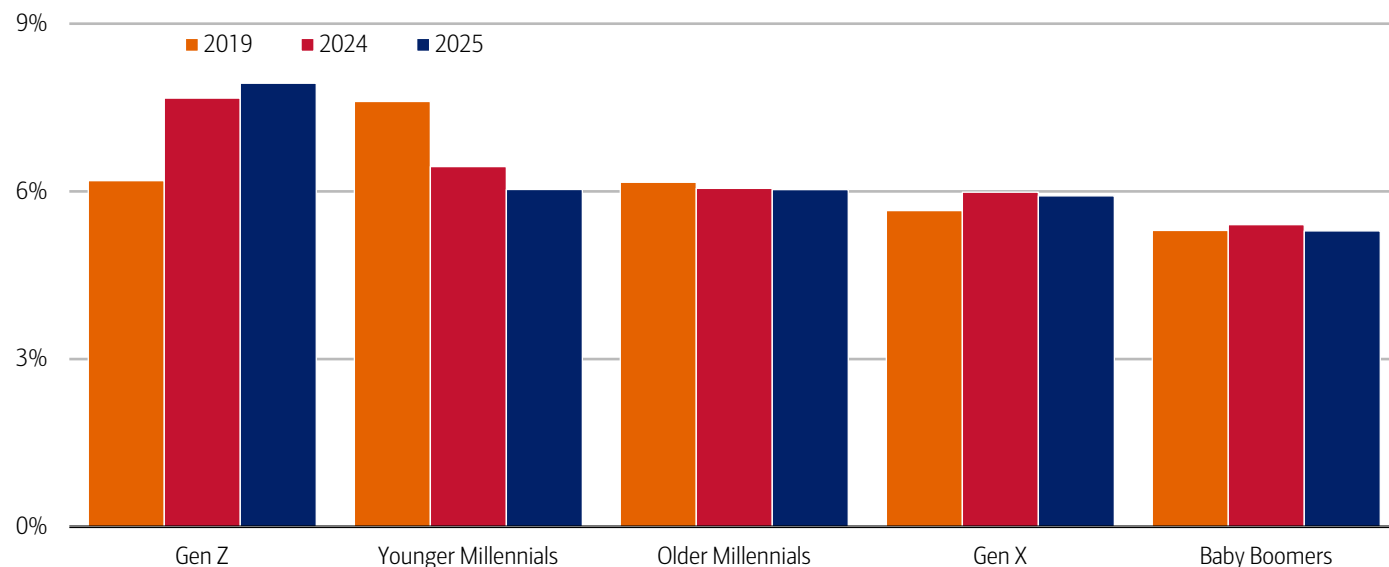
BANK OF AMERICA INSTITUTE

Passing the club: Millennials ease, as Gen Z drives adoption

Looking at golf spending by age, there's been a substantial drop in the share of younger Millennials participating in the sport, along with a smaller decrease among older Millennials since 2019 (Exhibit 3). Some in this generation may be pulling back as they take on more life responsibilities, such as starting families or building their careers. Other may be pulling back on discretionary spending as they took on more financial responsibilities during a period of elevated costs – driven by inflation and higher-than-usual interest rates (read more in [Autos: stuck in a lower gear?](#)).

Exhibit 3: There's been a significant increase in the share of Gen Z households spending on golf since the pandemic

Share of households with golf spending by age generation (yearly, %)



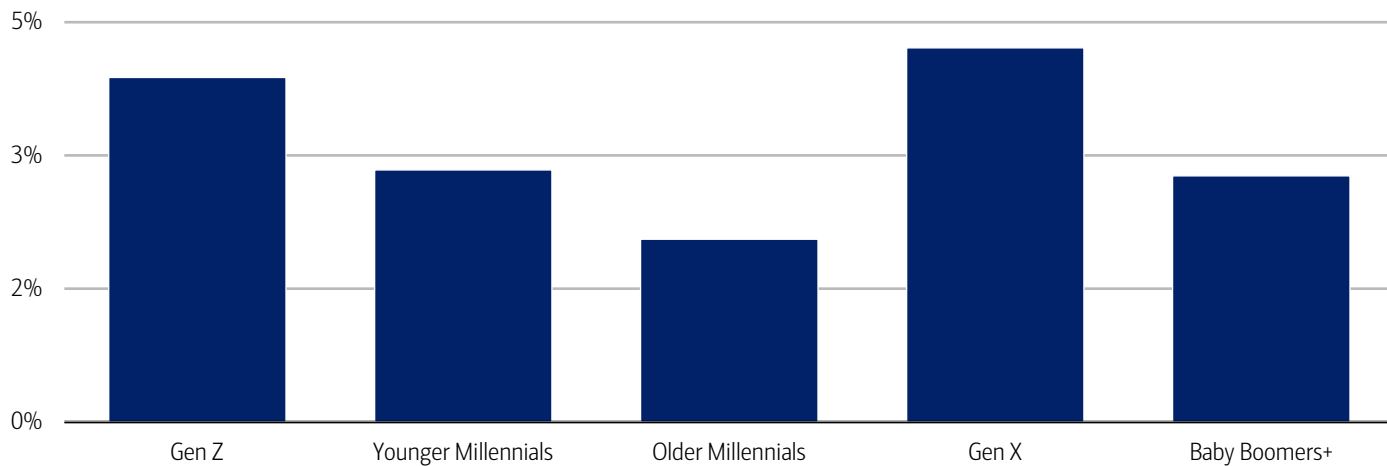
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Conversely, there's been a significant increase in the share of Gen Z households spending on golf. In our view, this could reflect the somewhat recent popularity of driving ranges and indoor simulators, as well as their desire to participate in more healthy experiences (read more in [Younger generations move from barstools to barbells](#)). It could also be that Gen Z came of age during a period of heightened social distancing, which may have encouraged adoption of outdoor sports like golf.

Furthermore, looking just at households that spent on golf in 2025, golf spending growth was faster among Gen Z but still slower than Gen X (Exhibit 4), likely reflecting the latter’s stronger overall financial position, allowing more spend and more time on the course.

Exhibit 4: Gen Z are catching up with Gen X in terms of fastest spending growth
Spending growth on golf by generation (2025 compared to 2024, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

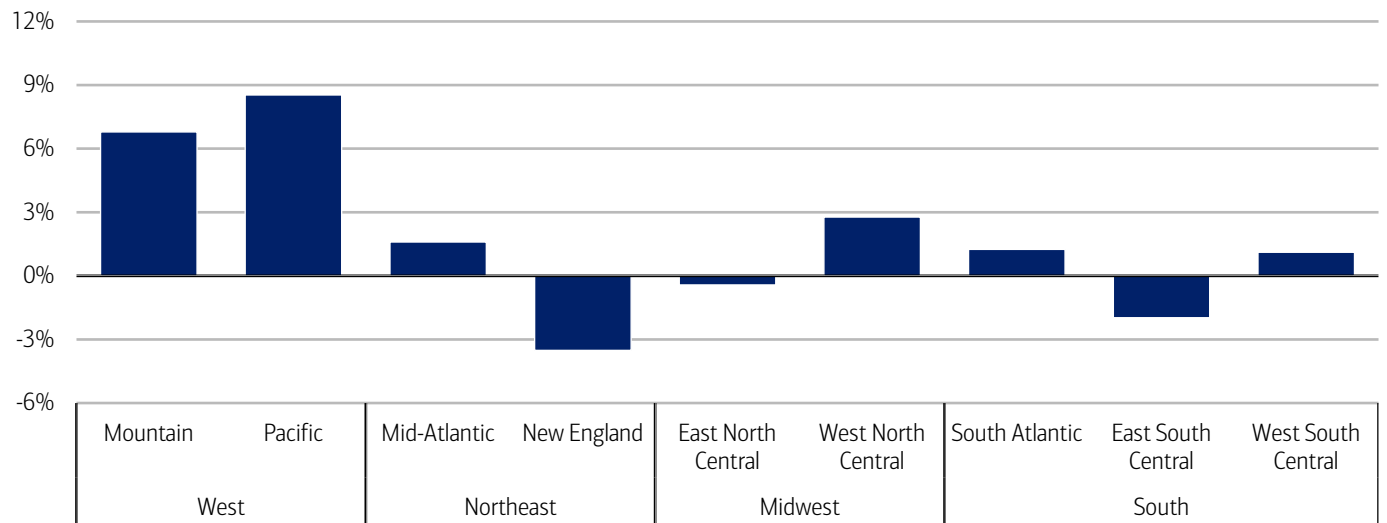
Golf heads West

Looking across the US, the West is clearly above par for golf. Spending increased around 7% and 9% YoY per customer in the Mountain (Arizona, Colorado, Utah – see Methodology for full list of states) and Pacific (California, Washington, etc.) divisions, respectively (Exhibit 5). The picture was mixed in all other regions.

In our view, more spending in the West may reflect some of the increases in domestic migration to the Western portion of the Sunbelt (read more in [On the move: US migration patterns](#)). Another part of the story may be the rise in “stay-to-play” policies, where golfers stay at the course and pay course fees. Also, some golf resorts in the region have added new courses that might incentivize some golfers to expand their length of stay and spend more money.

Other divisions including the West North Central (e.g., Minnesota, Nebraska, Missouri, etc.) and the West South Central (Texas, Louisiana, etc.) have experienced an increase in the number of residents playing golf, although new players are gravitating toward less expensive golfing experiences.

Exhibit 5: Golf spending in the West rose at least twice as fast YoY than any other region in 2025
Aggregate spending by US Census region and division (2025 compared to 2024, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under \$5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

Golf spending includes spending at courses, mini golf, driving ranges, and simulators but does not include country club dues or golf retailers.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation,

changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

US Census Regions of the United States:

Northeast: New Jersey, Pennsylvania, New York (Mid-Atlantic); Maine, Connecticut, Massachusetts, New Hampshire, Rhode Island, Vermont (New England)

Midwest: Indiana, Wisconsin, Illinois, , Michigan, Ohio (East North Central); Missouri, Minnesota, Iowa, Nebraska, North Dakota, South Dakota, Kansas

South: Delaware, Texas, Louisiana, Arkansas, Oklahoma (West South Central); Mississippi, Alabama, Tennessee, Kentucky (East South Central); Washington DC, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, West Virginia (South Atlantic)

West: Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, Wyoming (Mountain); Alaska, California, Hawaii, Oregon, Washington (Pacific)

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Joe Wadford

Economist, Bank of America Institute

Sources

Jon Kaplan

Senior Vice President, Consumer & Small Business Products and Analytics

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice. Copyright 2026 Bank of America Corporation. All rights reserved.

Transformation

Feeding the world with AI

07 April 2026

Key takeaways

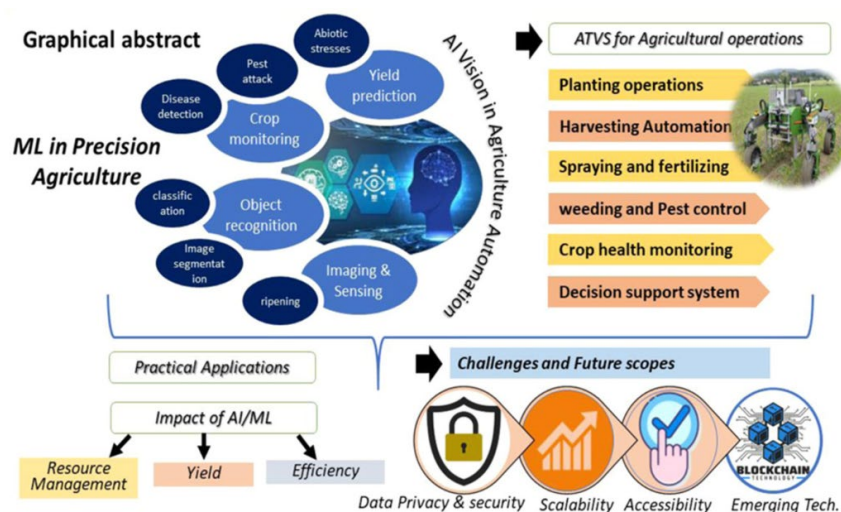
- Agriculture is undergoing its biggest technological shift in decades as AI becomes embedded across soil management, irrigation, fertilization and crop monitoring. By 2024, over half of farmers had adopted or were willing to adopt AI-enabled tools, driven by measurable gains in decision-making, yields, efficiency and sustainability.
- Digital AI has transformed how farmers understand their fields, but insight alone is no longer sufficient amid climate volatility, labor shortages, rising input costs and non-linear yield risks - including geopolitical instability along critical fertilizer supply corridors. These pressures demand not just better decisions, but precise, timely, plant-level action - something traditional advisory AI cannot deliver.
- That execution gap is pulling the sector toward physical AI, which enables real-time, plant-by-plant control. This shift from "AI for advice" to "AI for autonomous agronomy" directly moves revenue, cost and risk curves in a sector defined by tight margins and biological variability.

Agriculture's latest technological evolution

Agriculture, a sector that represents roughly 4% of global gross domestic product (GDP), is undergoing its biggest technological shift in decades, according to BofA Global Research. AI has already become deeply embedded in modern farming practices, improving farmers' decisions around soil, irrigation, fertilization, crop monitoring and disease detection (Exhibit 1). And the use of sensors, drones, satellite imaging and predictive crop models creates a rich layer of digital intelligence that can help farmers understand their fields with unprecedented clarity.

Exhibit 1: Crop monitoring + resource management = higher productivity for sustainable agriculture

Infographic illustrating machine learning (ML) in precision agriculture



Source: www.sciencedirect.com CC-by 4.0¹; BofA Global Research

Note: ATVs = all-terrain vehicles

BANK OF AMERICA INSTITUTE

¹ Kumar, A., Kumar, R., Padhiary, M., Saha, D., & Sethi, L.N. (2024, June 3). Enhancing precision agriculture: A comprehensive review of machine learning and AI vision applications in all-terrain vehicle for farm automation. ScienceDirect. <https://doi.org/10.1016/j.jatech.2024.100483>

AI moves from early adoption to sector-wide penetration

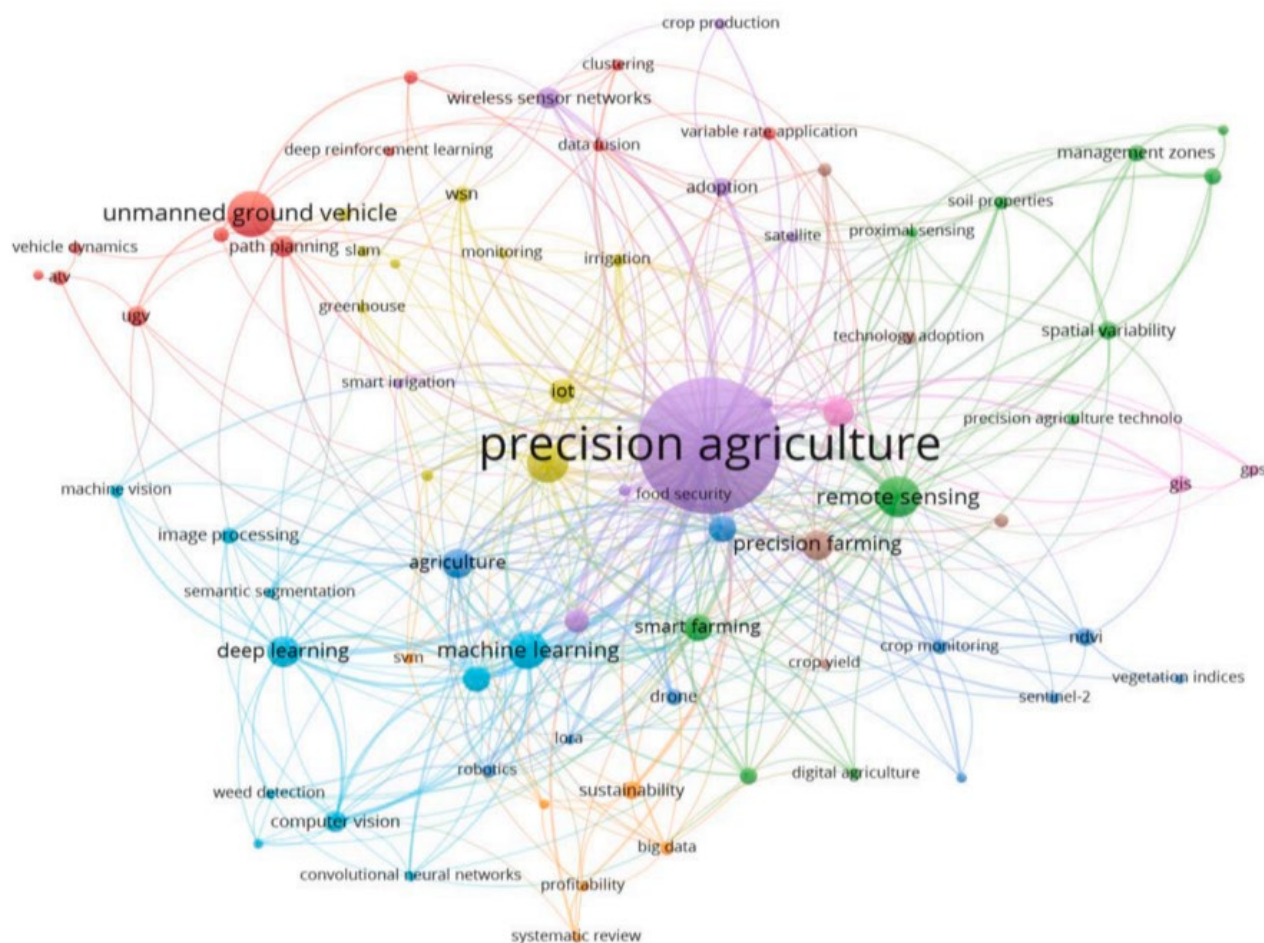
As of 2024, over half of farmers worldwide had adopted or were willing to adopt at least one precision-agriculture or AI-enabled technology.² This uptake is driven by tangible agronomic benefits (related to crop growing and soil management): 80% of farmers using data analytics report better decision-making. AI-enabled precision irrigation and fertilization can raise crop yields by 25%.³ Additionally, AI improves water usage efficiency and fertilizer application accuracy,⁴ while also reducing emissions⁵ – an increasingly important advantage when fertilizer supplies are constrained – thereby promoting sustainable farming practices.

AI-in-agriculture market: ~\$47 billion by 2034E

Applications of AI in agriculture now span an increasingly diverse set of technologies, from AI-guided vertical farming and biological enhancement tools, to AI-based crop modeling and precision agriculture using drones, sensors and satellites for real-time field intelligence. Money flows reflect this shift: agricultural technology (agtech) companies raised \$7 billion in 2025, up nearly 4% year-over-year (YoY), with precision-agriculture companies outpacing deals for crop inputs and enhancements.⁶

Exhibit 2: Deploying precision agriculture for farm automation is attracting growing interest from researchers and industry stakeholders

Keyword network diagram for precision agriculture



Source: www.sciencedirect.com CC-by 4.0, BofA Global Research

Note: The keyword network diagram based on 77 most occurring keywords out of 2998, with number of occurrences of a keyword being ≥ 5 (Based on 1055 publication accessed from Scopus database with keywords 'ML in agriculture, AI vision, ATV, Precision Agriculture and unmanned ground vehicle')(2024, March 17)

BANK OF AMERICA INSTITUTE

² Statista Research Department. (2024, October). *Share of farmers using or willing to adopt at least one new technology in the US in 2024, by technology type*. Statista.

³ Ferdusmou, J., Hossain, M.A., Khatoon, R., & Saha, S. (2025, January). *Smart Farming Revolution: AI-Powered Solutions for Sustainable Growth and Profit*. Journal of Management World.

⁴ Ibid.

⁵ Hoque, A., Padhiary, M., Roy, S., & Saikia, P. (2025, March 18). *Climate-Smart Agriculture: AI-Based Solutions for Enhancing Crop Resilience and Reducing Environmental Impact*. Asian Research Journal of Agriculture 18(1): 291-310. <https://doi.org/10.9734/arja/2025/v18i1665>

⁶ Frederick, A. (2026, January 6). *AgriFood VC First Look*. PitchBook.

⁷ Kumar, A., Kumar, R., Padhiary, M., Saha, D., & Sethi, L.N. (2024, June 3). *Enhancing precision agriculture: A comprehensive review of machine learning and AI vision applications in all-terrain vehicle for farm automation*. ScienceDirect. <https://doi.org/10.1016/j.atech.2024.100483>

Broader market outlooks underscore this transition. The AI-in-agriculture market is forecasted to increase at a 26.3% compound annual growth rate (CAGR) to \$46.6 billion by 2034, per Global Market Insights.⁸ This is driven by increased use of precision inputs, labor substitution and real-time agronomic decision support. Machine learning – now representing roughly half of the market – underpins emerging technologies such as generative AI, autonomous tractors and robotic sprayers (read [AI dictionary, part 1: The basics](#) for an intro to machine learning). It also enables real-time object recognition across key agricultural applications including weed detection, livestock monitoring and yield estimation from aerial imagery.

According to BofA Global Research, North America currently leads agricultural AI adoption with a 36% market share, supported by strong capex appetite and a mature precision-agriculture ecosystem that speeds up deployment of autonomous equipment. All these trends show that farming is shifting from *digital* agronomy (decision support) to *autonomous* agronomy (execution) (Exhibit 2). Furthermore, this expansion of AI-driven and robotics-based farming technology is supported by favorable regulation, which simplifies deployment. Falling hardware costs for sensors, compute and machine vision also make adoption more affordable. At the same time, policy incentives are increasingly aligned with technologies that verifiably reduce chemical inputs, accelerating uptake.

Physical AI: Brains behind the wheel

Physical AI integrates artificial intelligence into physical systems, such as humanoid robots, autonomous vehicles and drones (we further define this topic in [Physical AI, part 1: The basics](#)). As a result, AI can perceive, reason and act in the real world – not just generate digital outputs. This matters because the next productivity wave won't come from better dashboards, but from execution at the edge, delivered by robots, drones, autonomous implements and sensor-driven machines that can continuously close the loop between sensing and action, per BofA Global Research.

Capital and capability are accelerating

Over 70 companies are now developing physical AI infrastructure across data, simulation, foundation models and observability. Robotics and physical AI companies raised ~\$41 billion in 2025, while equity deals into robot foundation model developers alone increased more than tenfold from three in 2021 to 32 in 2025. Collectively, these advances position physical AI as a shared, reusable infrastructure, deployed across robots, vehicles and drones, rather than an isolated, task-specific solution.

The shift is structural: Climate volatility is tightening constraints of farming

According to BofA Global Research, agriculture is a sector where physical AI can scale in the near term because the underlying operating environment has changed. Fertilizer markets are tightly locked into a handful of regions and energy routes, so even localized geopolitical disruptions can quickly ripple through global input costs.

And on the climate front, drought and heat stress are no longer episodic; they are persistent conditions that directly cap yields and destabilize broader farm economics (Exhibit 3). The European Union's Joint Research Centre found that 52 individual prolonged meteorological drought events occurred from August 2023 to July 2024.⁹ Droughts, heatwaves and warm spells have negatively impacted crop productivity across Europe, Southern Africa, Central and Southern America and Southeast Asia. In this regime, many crops exhibit non-linear responses to temperature. Yields can improve gradually up to a threshold, then deteriorate sharply once exceeded. Meanwhile, rising heat accelerates evapotranspiration (loss of water to the atmosphere from soil evaporation and plant transpiration), pulling crops into water stress faster and worsening pest and disease pressure.

Why “advisory AI” hits a ceiling

Digital AI has transformed how farmers understand their fields, but insight alone is no longer enough. The pressures reshaping global agriculture are climate volatility, chronic labor shortages, rising input costs and non-linear yield risks. These require not just better decisions, but precise, timely, plant-level action. This is where traditional “advisory” AI reaches its ceiling. A farmer can know a disease outbreak is emerging, yet without the labor, equipment or precision to intervene quickly and locally, the insight can't be monetized. Modern agriculture now needs systems that can see, decide and act in the real world. That execution gap is what is pulling the sector toward physical AI.

Physical AI addresses the gap by enabling real-time, plant-by-plant control – dynamic irrigation, targeted nutrient delivery, early pest intervention and precision application based on live perception. This marks a shift from “AI for advice” to “AI for autonomous agronomy.” In a sector defined by narrow margins and biological variability, that shift directly moves revenue, cost and risk curves.

Economics is driving commercial scale

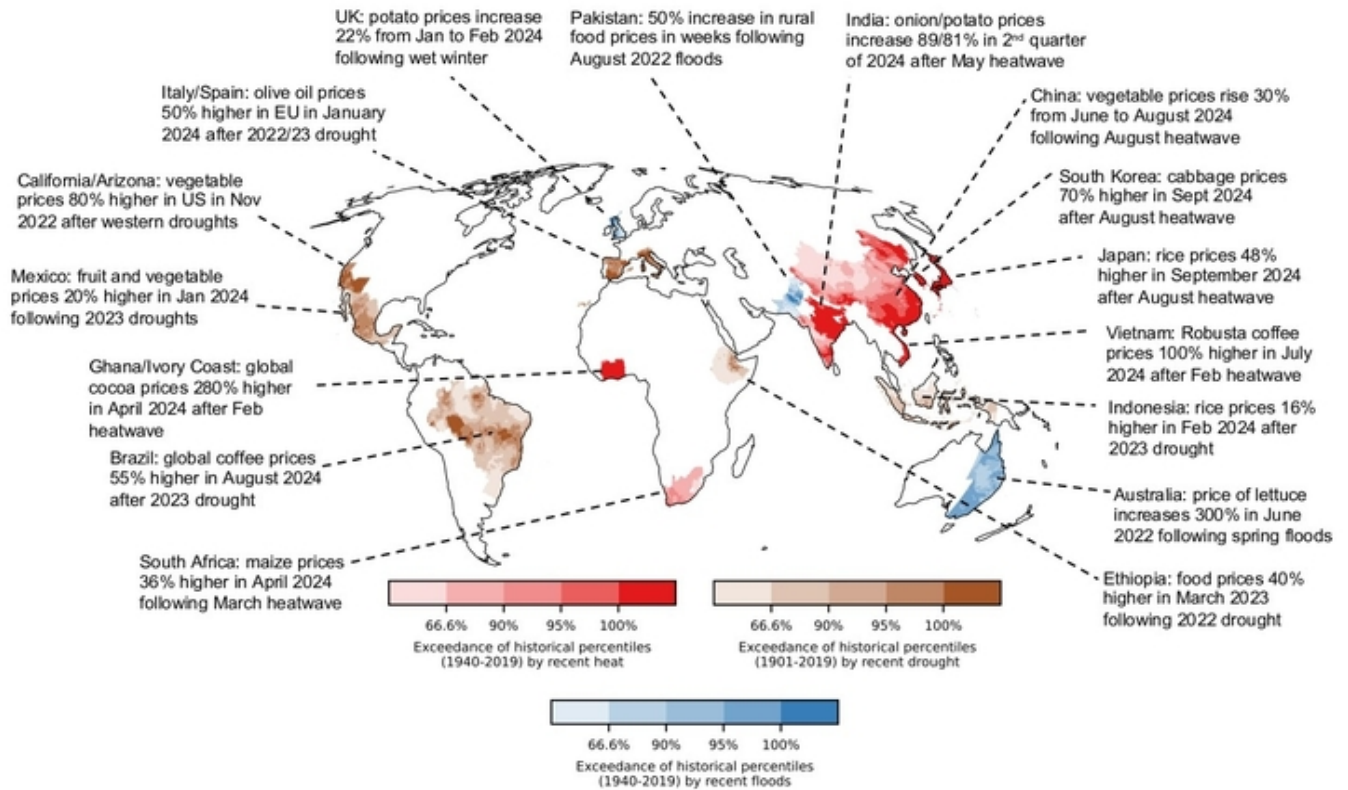
Physical AI is scaling because the economics have turned. Technologies that deliver payback in under a year scale fastest in agriculture and falling hardware costs are strengthening this equation. Precision robotics that reduce labor, chemical use and operational time can pay back in months rather than years. Yield stability in a more volatile climate becomes an additional economic engine. The shift is from automation as a way to reduce costs to a means to boost profitability and resilience.

⁸ Global Market Insights. (2025, May). *AI in Agriculture Market Size & Share 2025-2034*.

⁹ Joint Research Centre. (2024, October 2). *Global drought threatens food supplies and energy production*. European Commission.

Exhibit 3: Food prices have spiked across the globe in response to extreme climate conditions

Map of climate-induced food price spikes since 2022



Source: ResearchGate CC-by 4.0¹⁰, BofA Global Research

Note: Underlying shading represents the extent to which the associated climate conditions exceeded percentiles of the historical distribution.

BANK OF AMERICA INSTITUTE

¹⁰ Donat, M.G., Kotz, M., Lancaster, T., Parker, M., Smith, P., Taylor, A., & Vetter, S. (2025, July). *Climate extremes, food price spikes, and their wider societal risks*. Environmental Research Letters. <https://doi.org/10.1088/1748-9326/ade45f>

Contributors

Lynelle Huskey

Analyst, Bank of America Institute

Vanessa Cook

Content Strategist, Bank of America Institute

Sources

Menka Bajaj

Sustainability Strategist, BofA Global Research

Haim Israel

Equity Strategist, BofA Global Research

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice. Copyright 2026 Bank of America Corporation. All rights reserved.